Sahaj Bhadja

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Summary

Junior software developer with 2+ years building **C++/Python applications**, simulation and data systems, and cross-platform software; experienced in component testing, functional verification, requirements analysis, technical writing, SQL Server, debugging, and automated delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Developed **Python application components and SQL Server workflows** for 500+ users, implementing normalized schemas, indexed access, and structured state handling to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Performed functional verification across 5,000+ monthly data workflows, designing validation, reconciliation, error-handling, and failure-isolation checks while documenting defects and releasing tested corrections through CI/CD with 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented **modular Python and JavaScript application components** for upload, processing, history, and reporting functions, supporting 150+ users and 10,000+ daily requests through maintainable Flask MVC services and React interfaces.
- Executed component and functional tests across PDF, DOCX, image, and ZIP inputs, reproducing defects and implementing secure validation, fallback handling, persistent results, and fault-tolerant multithreaded processing while reducing turnaround 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **simulation and evaluation software** over 10,000 autonomous-driving episodes, integrating map geometry and temporal signals to improve long-horizon trajectory planning 18% and closed-loop behavioral stability 15%.
- Defined test scenarios, analyzed performance metrics and failure patterns, and documented system constraints for reproducible verification, collaborating with researchers to refine algorithms and improve functional behavior across simulated operating conditions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed and refactored **object-oriented cross-platform software components** using Singleton and Factory patterns, APIs, SQL, and caching, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability.
- Automated component builds, installation, and **20+ unit and integration tests** across Windows, Linux, and macOS, generating coverage summaries and failure reports while delivering controlled, versioned releases through Jenkins.

Pandit Deendayal Energy University **India**
Research Fellow *January 2023 – May 2024*
- Analyzed existing distributed-system protocols and developed a **new consensus algorithm** integrating 5+ lattice-based cryptographic primitives, reducing centralized-control and quantum-risk exposure 25% through structured benchmark simulations and analysis.
- Produced detailed **technical writing and verification documentation** for algorithm design, assumptions, benchmarks, and security behavior, contributing to one peer-reviewed IEEE paper and presenting findings at IISc Bengaluru.

Skills

Languages: C++, C, Python, Java, SQL, JavaScript, TypeScript, Bash
Microsoft & Web: SQL Server, Azure SQL, Windows, Angular.js, React.js, HTML, CSS, REST APIs, WebSockets
Software Engineering: OOP, design patterns, data structures, algorithms, multiprocess concepts, system integration
Testing: Component testing, functional testing, unit/integration testing, validation, verification, PyTest, Mockito, Selenium
Delivery: Git, Jenkins, GitHub Actions, CI/CD, Docker, Linux, requirements analysis, debugging, technical documentation

Projects

CPU Scheduling Simulator Live | *C++, Algorithms, Functional Testing*
- Implemented and functionally tested **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.

Chat2SQL Code | *Java, TypeScript, SQL Server, Docker, Maven*
- Enhanced a **cross-platform database application** across Java and TypeScript components, supporting SQL Server and 10+ SQL/NoSQL systems with schema exploration, structured query execution, Docker, Maven, and reusable development workflows.

Linguistix | *Python, AWS, Docker, CI/CD*
- Developed and deployed a **containerized Python processing component** on AWS ECS/Fargate with CodePipeline, automated validation checks, and repeatable CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.